Week 12 Reading: Weather types and downscaling

Downscaling climate model rainfall through weather types

2026-11-09

> Draft. This page is unfinished and will change.

Assigned

1. Moron et al. (2008), §1, §3a-c, §4a, §5b-c, and §6.

Background

1. Section 3.5.2 of Ghil et al. (2011), the paragraphs on weather typing and the analog method.

Further reading

1. Muñoz et al. (2017), Eq. 10 and Fig. 12.

Bibliography

Ghil, M., Yiou, P., Hallegatte, S., Malamud, B. D., Naveau, P., Soloviev, A., et al. (2011). Extreme Events: Dynamics, Statistics and Prediction. *Nonlinear Processes in Geophysics*, *18*(3), 295–350. https://doi.org/10/fvzxvv

Moron, V., Robertson, A. W., Ward, M. N., & Ndiaye, O. (2008). Weather Types and Rainfall over Senegal. Part II: Downscaling of GCM Simulations. *Journal of Climate*, *21*(2), 288–307. https://doi.org/10.1175/2007JCLI1624.1

Muñoz, Á. G., Yang, X., Vecchi, G. A., Robertson, A. W., & Cooke, W. F. (2017). A Weather-Type-Based Cross-Time-Scale Diagnostic Framework for Coupled Circulation Models. *Journal of Climate*, *30*(22), 8951–8972. https://doi.org/10.1175/jcli-d-17-0115.1